

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1691

Parsons Tap 138 kV
March 25, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1691. The Interconnection Customer proposes a 202.3 MW solar photovoltaic facility interconnecting at the Parsons Tap 138 kV substation in the Virginia area. The request was received on March 09, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 593 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 24 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,655,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: N Randolph to Yellow Crk 138kV
Contingency: Haggin to N Tower Tap 345kV
Loading: 112.8% (Rating: 241 MVA)
DFAX: 5.0%
Pre-project loading: 108.6%
2. Facility: Spring St to Wing Lane 138kV
Contingency: Mitchell Dam to Skaggs Island Junction 2 345kV
Loading: 112.7% (Rating: 309 MVA)
DFAX: 8.0%
Pre-project loading: 107.5%
3. Facility: Deltaport to Potosi 138kV
Contingency: Switzer to Pheasant Branch 138kV
Loading: 110.9% (Rating: 365 MVA)
DFAX: 5.3%
Pre-project loading: 108.0%
4. Facility: Haggin Substation 345 kV/138 kV Auto
Contingency: N Tower Tap to Haggin 345kV
Loading: 110.6% (Rating: 421 MVA)
DFAX: 6.3%
Pre-project loading: 107.5%
5. Facility: Wing Lane to Seco 138kV
Contingency: Mitchell Dam to Skaggs Island Jct 2 345kV
Loading: 110.3% (Rating: 413 MVA)
DFAX: 10.7%
Pre-project loading: 105.1%

6. Facility: Potosi to Tyler 138kV
Contingency: Switzer to Pheasant Branch 138kV line
Loading: 110.1% (Rating: 396 MVA)
DFAX: 5.3%
Pre-project loading: 107.4%
7. Facility: N Randolph to Bairoa 3008 138kV
Contingency: North Tower Tap to Haggin 345kV
Loading: 108.8% (Rating: 237 MVA)
DFAX: 5.0%
Pre-project loading: 104.6%
8. Facility: Seco to Harvard 138kV
Contingency: Mitchell Dam to Skaggs Island Jct 2 345kV line
Loading: 108.7% (Rating: 524 MVA)
DFAX: 10.7%
Pre-project loading: 104.6%
9. Facility: Hull St Road Substation to Harvard Substation 138kV Line
Contingency: Mitchell Dam to Skaggs Island Jct 2 345kV line
Loading: 108.2% (Rating: 574 MVA)
DFAX: 10.7%
Pre-project loading: 104.4%
10. Facility: Redbed Pains Wind Farm to Westcoast Cellufibre 138kV
Contingency: Skaggs Island Jct 2 to Mitchell Dam 345kV
Loading: 106.0% (Rating: 509 MVA)
DFAX: 6.8%
Pre-project loading: 103.3%
11. Facility: Fruitridge & S. Land Park to Redbed Plains Wind Farm 138kV Line
Contingency: Mitchell Dam to Skaggs Island Junction 2 345kV
Loading: 105.9% (Rating: 341 MVA)
DFAX: 6.8%
Pre-project loading: 101.9%
12. Facility: Carlin to Fruitridge & S. Land Park 138kV
Contingency: Mitchell Dam to Skaggs Island Junction 2 345kV
Loading: 105.5% (Rating: 385 MVA)
DFAX: 6.8%
Pre-project loading: 102.0%
13. Facility: Deltaport to Moonstone 138kV
Contingency: Pheasant Ranch to Switzer Substation 138kV
Loading: 104.0% (Rating: 352 MVA)
DFAX: 5.3%
Pre-project loading: 100.9%
14. Facility: Chancellor Row to Bear Island 345kV
Contingency: Leavitt to Birchwood 345kV
Loading: 102.5% (Rating: 2,328 MVA)
DFAX: 24.7%
Pre-project loading: 100.3%
15. Facility: Plaza to Big Hanaford 138kV
Contingency: Skaggs Island Jct 2 to Mitchell Dam Substation 345kV
Loading: 102.1% (Rating: 716 MVA)
DFAX: 6.2%
Pre-project loading: 100.4%
16. Facility: Spring St to Chipola Substation 138kV
Contingency: Mitchell Dam to Skaggs Island Jct 2 Substation 345kV line

- Loading: 102.1% (Rating: 300 MVA)
DFAX: 8.0%
Pre-project loading: 96.7%
17. Facility: Bobville to Cochise 138kV
Contingency: Yarnen to McCartney 345kV
Loading: 101.6% (Rating: 317 MVA)
DFAX: 10.7%
Pre-project loading: 94.7%
18. Facility: Mickens to Igo 345kV Line
Contingency: Mickens to Newport News 345kV
Loading: 101.4% (Rating: 1,089 MVA)
DFAX: 5.5%
Pre-project loading: 100.4%
19. Facility: Leavitt to Birchwood 345kV
Contingency: Bear Island to Dunn Siding 345kV L
Loading: 101.3% (Rating: 3,031 MVA)
DFAX: 24.5%
Pre-project loading: 99.6%
20. Facility: Big Hanaford to Carver 138kV
Contingency: Plaza to Big Hanaford 138kV L
Loading: 101.1% (Rating: 495 MVA)
DFAX: 5.4%
Pre-project loading: 98.9%
21. Facility: Haggin to Mitchell Dam 345kV
Contingency: Leavitt to Chancellor Row Substation 345kV
Loading: 100.7% (Rating: 1,658 MVA)
DFAX: 23.0%
Pre-project loading: 97.9%
22. Facility: Dg Amp Solar Coldwater to Kigs Riv Substation 138kV
Contingency: Mitchell Dam to Skaggs Island Jct 2 345kV
Loading: 100.4% (Rating: 435 MVA)
DFAX: 5.9%
Pre-project loading: 97.6%
23. Facility: Dunn Siding to Birchwood 345kV L
Contingency: Leavitt to Birchwood 345kV line
Loading: 100.3% (Rating: 1,519 MVA)
DFAX: 7.6%
Pre-project loading: 99.3%
24. Facility: Carver to Shockoe 138kV line
Contingency: Plaza to Big Hanaford 138kV line
Loading: 100.1% (Rating: 574 MVA)
DFAX: 5.4%
Pre-project loading: 98.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N Randolph to Yellow Crk 138kV	Rebuild line with higher-capacity conductor	\$7,045,000	6% (\$433,000)
Spring T to Wing Lane 138kV	Rebuild line with higher-capacity conductor	\$1,439,000	4% (\$65,000)
Deltaport to Potosi 138kV	Rebuild line with higher-capacity conductor	\$7,401,000	4% (\$293,000)
Haggin Substation 345 kV/138 kV XFMR	Replace transformer with higher-rated unit	\$11,232,000	4% (\$396,000)
Wing Lane to Seco 138kV	Rebuild line with higher-capacity conductor	\$2,789,000	4% (\$106,000)
Potosi to Tyler 138kV	Rebuild line with higher-capacity conductor	\$537,000	4% (\$21,000)
North Randolph to Bairoa 3008 138kV	Reconductor line	\$3,024,000	6% (\$186,000)
Seco to Harvard 138kV	Reconductor line	\$1,844,000	4% (\$70,000)
Hull Sreet Rd to Harvard 138kV	Reconductor line	\$433,000	4% (\$16,000)
Redbed Pains Wind Farm to Westcoast Cellufibre 138kV	Reconductor line	\$1,396,000	3% (\$36,000)
Redbed Plains Wind Farm to Fruitridge & S. Land Park 138kV	Reconductor line	\$3,759,000	4% (\$131,000)
Carlin to Fruitridge & S. Land Park 138kV	Reconductor line	\$500,000	4% (\$17,000)
Deltaport to Moonstone 138kV	Reconductor line	\$508,000	4% (\$20,000)
Bear Islad to Chancellor Row 345kV	Reconductor line	\$10,544,000	2% (\$200,000)
Plaza to Big Hanaford 138kV	Reconductor line	\$2,313,000	2% (\$58,000)
Sping St to Chipola 138kV	Reconductor line	\$4,010,000	4% (\$181,000)
Cochise to Bobville 138kV	Reconductor line	\$5,426,000	9% (\$465,000)
Mickens to Igo 345kV Line	Reconductor line	\$20,920,000	2% (\$439,000)
Leavitt to Birchwood 345kV	Reconductor line	\$55,072,000	2% (\$1,003,000)
Carver to Big Hanaford 138kV L	Reconductor line	\$672,000	3% (\$20,000)
Haggin to Mitchell Dam 345kV	Reconductor line	\$7,911,000	3% (\$204,000)
Dg Amp Solar Coldwater to Kigs Riv Substation 138kV	Reconductor line	\$927,000	3% (\$24,000)
Dunn Siding to Birchwood 345kV	Reconductor line	\$15,202,000	2% (\$259,000)
Carver to Shockoe 138kV	Reconductor line	\$393,000	3% (\$12,000)
	Total allocated to project		\$4,655,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.